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Short Notes

PHYSICS

- Created By **100%ilers** & Ajay Bhaiya
- Revise **01 Chapter** in just in **4 Minutes!**
- Special **Advanced** section from Notes of Top Rankers

100 Percentiler in PHYSICS

PHYSICS

CLASS XII

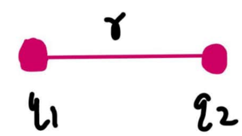
UNIT 1

ELECTROSTATICS

Coulomb's Law

Force between two point charges q_1 & q_2 is directly proportional to the product of charges and inversely proportional to the square of distance between them.

$$F \propto \frac{q_1 q_2}{r^2} = \frac{k q_1 q_2}{r^2} \quad \text{where} \quad k = \frac{1}{4\pi\varepsilon_0}$$



$$\boxed{k = 9\times10^9\ C^2/Nm^2} \qquad \boxed{\varepsilon_0 = 8.85\times10^{-12}}$$

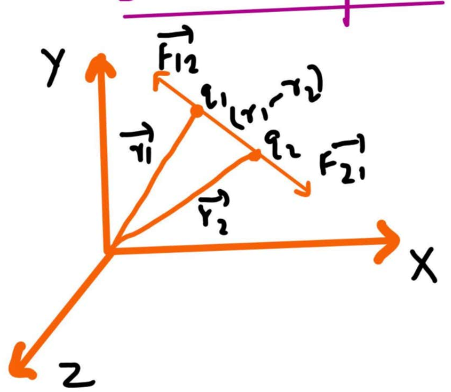
Force in medium

$$\boxed{F_m = \frac{1}{4\pi\varepsilon_r}\frac{q_1 q_2}{r^2}} \qquad \boxed{\frac{F}{F_m} = \uparrow\varepsilon_r} \qquad \boxed{\varepsilon_r = \frac{\varepsilon}{\varepsilon_0}}$$

Force if water is medium

$$\boxed{F_m = \frac{F}{80}}$$

In vector form



$$\vec{F}_{21} = -\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}$$

$$= -\frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^3}\vec{r}$$

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}\hat{r}$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^3}\vec{r}$$

Forces between multiple charges :

$$F_{1n} = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{\infty} \frac{q}{r_i^2} \hat{r}_1$$

Continuous Charge Distribution

1) Linear Charge Density $\quad \lambda = \frac{Q}{I}\ c/m$

2) Surface Charge Density $\quad \sigma = \frac{q}{A}\ c/m^2$

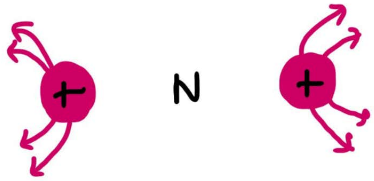
3) Volume charge Density $\quad \rho = \frac{Q}{V}\ c/m^3$

4) Electric field Density $\quad \vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\varepsilon_0} \frac{q}{r^2}$

5) Principle of Superposition $\quad \vec{E} = \sum_{i=1}^{nA} \vec{E}_1$

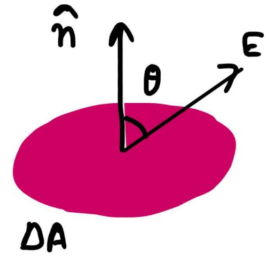
where $\vec{E}_1 = \frac{1}{4\pi\varepsilon_0} \frac{q_i}{r_i^3} \vec{r}_1$

Electric Field Lines



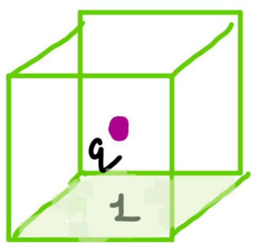
$\vec{E}_{AN} = \vec{E}_{BN} \Rightarrow \frac{q}{AN^2} : \frac{q}{BN^2}$

Electric Flux



$\Delta\phi_{net} = \vec{E}\cdot\vec{A}$

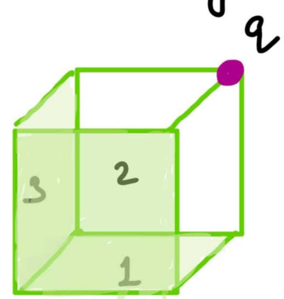
If +q is placed at centre of the cube



$\Delta\phi_{net} = \frac{q}{\varepsilon_0}$

$\Delta\phi_1 = \frac{q}{6\varepsilon_0}$

If +q is placed at one edge



$\Delta\phi_{edge}^{any} = \frac{q}{8\varepsilon_0}$

$\Delta\phi_{123} = \frac{q}{24\varepsilon_0}$

Gauss's Law

The total flux linked with a closed surface is $1/\varepsilon_0$ times the charge enclosed by the closed surface.

$$\boxed{\oint \vec{E}.d\vec{s} = \frac{q}{\varepsilon_0}}$$

Electric Potential

$V = -\frac{W_E}{q_0}$ J/C

Electric Potential Difference

$W_E = -q_0(V_B - V_A)$

Relation b/w E and V :

(i) $\vec{E} = -\frac{dv}{dr}$

(ii) $V_A - V_B = Ed$

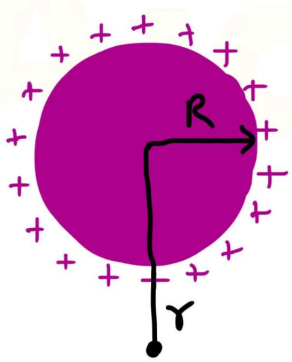
V due to multiple charges:

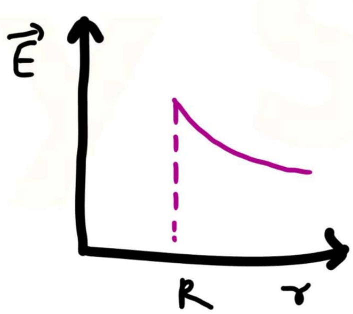
$$V = K \sum_{i=1}^{i=n} \frac{q_i}{r_{ip}}$$

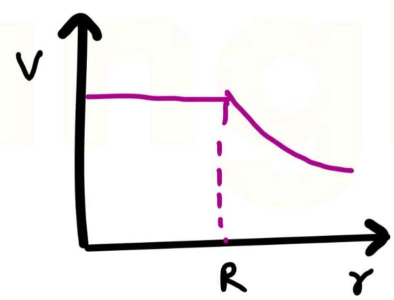
$r_{ip} \Rightarrow$ position vector

Hollow Sphere / Hollow shell / Spherical Conductor

	In	Out	Surface
$\vec{E}$	0	$\frac{kQ}{r^2}$	$\frac{kQ}{R}$
V	$\frac{kQ}{R}$	$\frac{kQ}{r}$	$\frac{kQ}{R}$

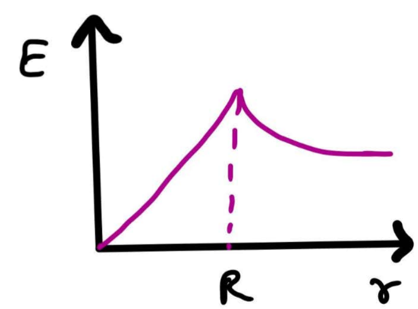


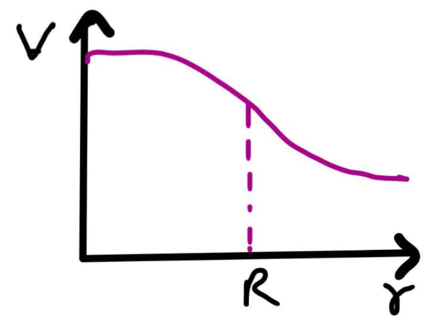




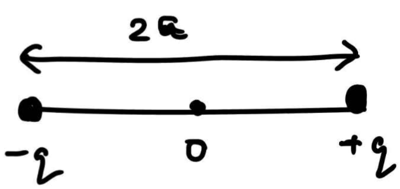
Solid Sphere / Solid Ball / Conductor

	Out	In	Centre	Surface
E	$\frac{kQ}{r^2}$	$\frac{kQr}{R^3}$	0	$\frac{kQ}{R^2}$
V	$\frac{kQ}{r}$	$\frac{kQ(3R^2-r^2)}{2R^3}$	$\frac{3kQ}{2R}$	$\frac{kQ}{R}$





Dipole



Dipole Moment

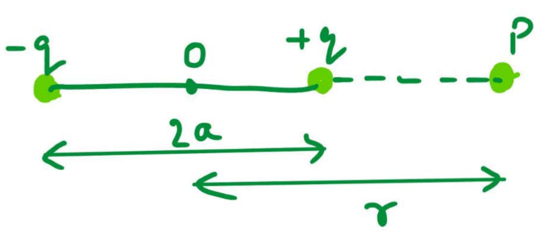
$$\boxed{\vec{P} = q(2a)\,\vec{p}}$$

On axial line

$$E = \frac{k2\vec{P}r}{(r^2-a^2)^2}$$

if $r^2 >>> a$ $\quad \vec{E}_{eq} = \frac{2k\vec{P}}{r^3}$

$$V = \frac{k\vec{P}\cdot\vec{r}}{r^3}$$

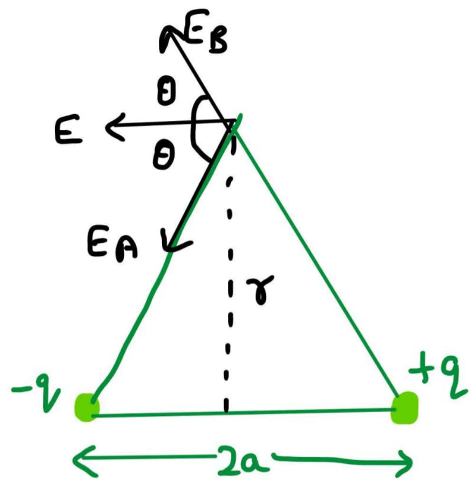


On equitorial plane

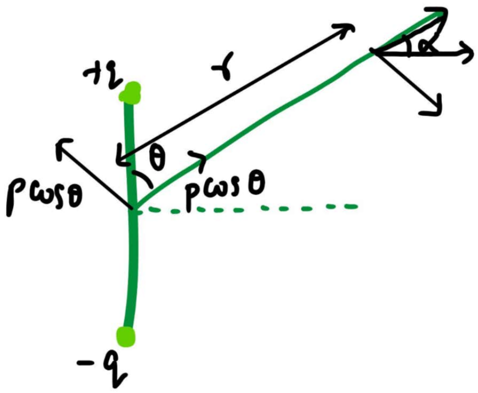
$$E_{eq} = \frac{-k\vec{P}}{(r^2+a^2)^{3/2}}$$

if $r >>> a$ $\quad E_{eq} = \frac{-k\vec{P}}{r^3}$

$$V = 0$$



At any arbitrary point

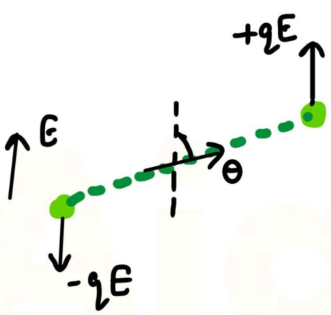


$$\vec{E}_{net} = \frac{kp}{r^3}\sqrt{1+\cos\theta}$$

$$\tan\alpha = \frac{1}{2}\tan\theta$$

$$V = \frac{kp\cos\theta}{r^2}$$

Torque on Dipole in a uniform External field



$$\boxed{\tau = Ep\sin\theta = p \times E}$$

Note: suspended dipole align itself to $\vec{E}$

θ gives angular displacement of dipole.

$\tau = pE\sin\theta$, if θ is small.

$\sin\theta \simeq \theta \quad \therefore \tau = -pE\theta$

Potential Energy of a dipole in a uniform Electric field

$U = -pE\cos\theta$

$U = -pE$ for $\theta = 0°$

$U = pE$ for $\theta = 180°$ (stable eqn)

(unstable eqn) (Max energy)

$$\boxed{\alpha = -\left(\frac{pE}{I}\right)} \qquad \boxed{\omega = \sqrt{\frac{pE}{I}}} \qquad \boxed{T = \sqrt{\frac{I}{pE}}}$$

Capacitance

Combination of Conductors

$Q \propto V$ or $\boxed{Q = CV}$

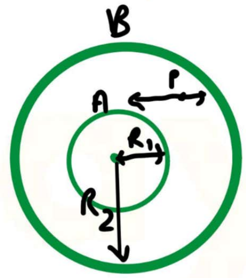
Capacitance of Spherical Conductors

1) Isolated sphere

$C = 4\pi\varepsilon_0 R$ or $C = \dfrac{R}{9\times10^9}$

$C_{earth} = 4\pi\mu F$

2) Concentric spherical shell

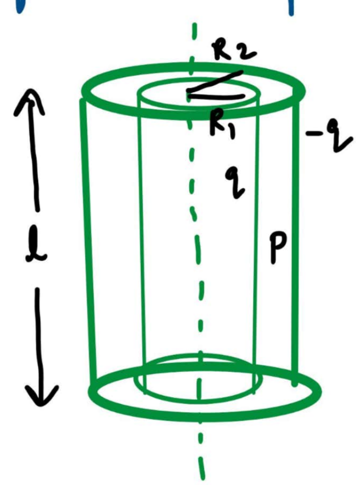


$V_p = KQ\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$

$E_p = \dfrac{kQ}{r^2}$

$C = \dfrac{4\pi\varepsilon_0 R_1 R_2}{R_2 - R_1}$

3) Cylindrical Capacitance

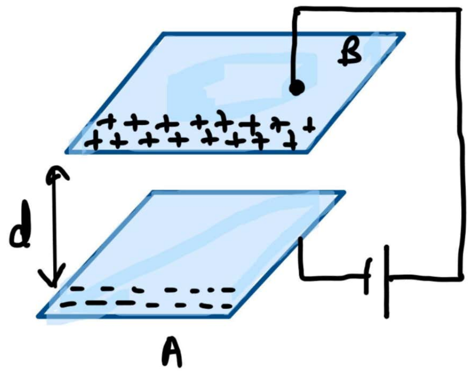


$V_p = \dfrac{2}{2\pi\varepsilon_0 l} \ln \dfrac{R_2}{R_1}$

$E_p = \dfrac{kQ}{r^2}$

$C = \dfrac{2\pi\varepsilon_0 l}{\ln\left(\dfrac{R_2}{R_1}\right)}$

Parallel Plate Capacitor



$$\boxed{E = \frac{Q}{A\varepsilon_0}}$$

$$\boxed{V = \frac{Qd}{\varepsilon_0 A}}$$

$$\boxed{C = \frac{k\varepsilon_0 d}{A}}$$

Effect of Dielectric on Parallel Plate Capacitor

$E_0 > E$ $\qquad$ $V_0 > V$ $\qquad$ $C > C_0$

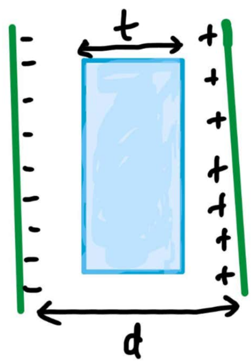
Polorisation

$P = \chi \varepsilon_0 E$

Susceptibility

$\chi = k - 1$

Dieleletric of thickness 't' is inserted



$$\boxed{V = \frac{Q}{\varepsilon_0 A}\left(d - t + \frac{t}{k}\right)}$$

k is dielectric constant

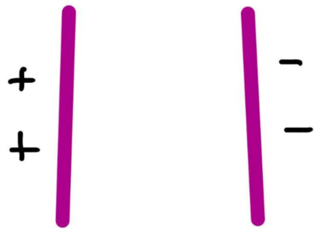
$$\boxed{C = \frac{\varepsilon_0 A}{d - t + t/k}}$$

Energy

✓ Electric field energy Density $= \frac{1}{2}k\varepsilon_0 E^2$ $\qquad$ for air $k = 1$

✓ Electric energy stored by a capacitor $=$ $U = \frac{1}{2}CV^2 = \frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C}$

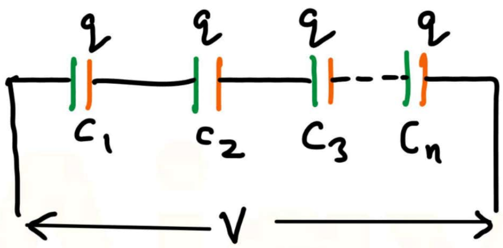
- If two different charges are on plate of capacitors



$$x = \frac{q_1 - q_2}{2}$$

Combination of Capacitors

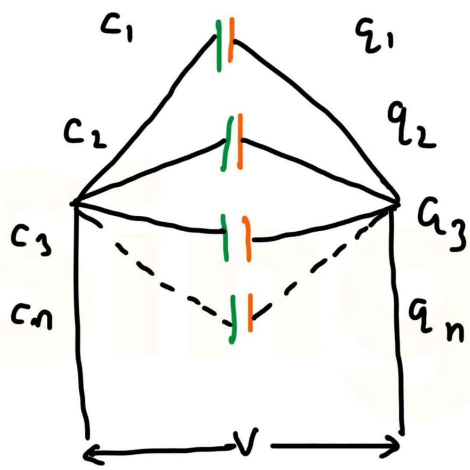
Series



$$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \text{----}$$

for 2 capacitors $\quad C = \dfrac{C_1 C_2}{C_1 + C_2}$

Parallel



$$E_{Cp} = C_1 + C_2 + C_3$$

for $2C \Rightarrow C_1 + C_2$

Sharing of charges in parallel capacitors.

★ Like plates are combined

$$\boxed{V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2} = \frac{q_1 + q_2}{C_1 + C_2}}$$

Heat loss $= \dfrac{1}{2} \dfrac{C_1 C_2}{C_1 + C_2} (V_1 - V_2)^2$

* Unlike plates are combined

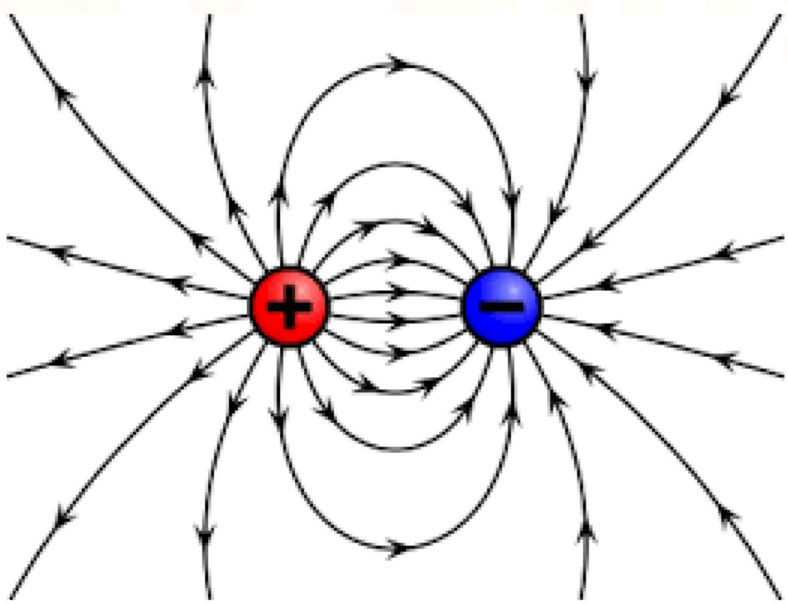
$$\boxed{V = \frac{C_1V_1 - C_2V_2}{C_1+C_2} = \frac{q_1 - q_2}{C_1+C_2}}$$

Heat loss $= \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_1+V_2)^2$

* If one capacitor is uncharged

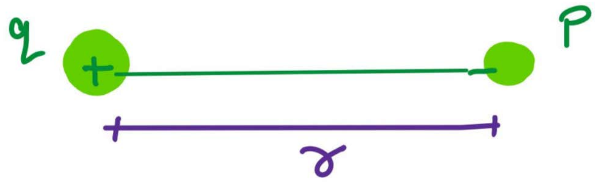
$$\boxed{V = \frac{C_1V_1}{C_1+C_2} = \frac{q}{C_1+C_2}}$$

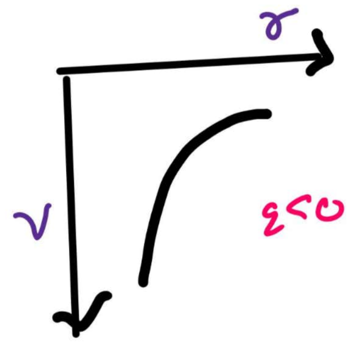
$$\Delta H = \frac{1}{2}\frac{C_1C_2}{C_1+C_2}(V_1)^2$$



Some important cases of electric field and potential. :

case 1: Point charge

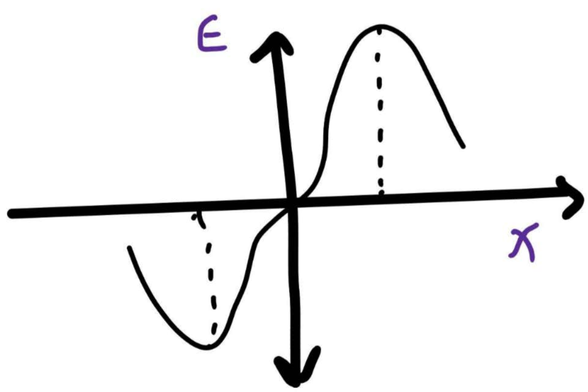


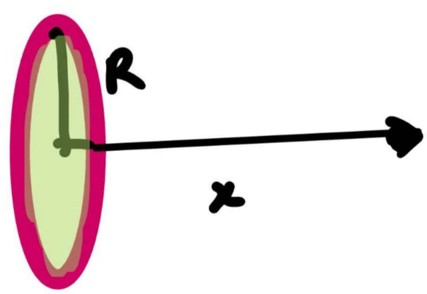


$$\boxed{\vec{E} = \frac{kq}{r^2}\hat{r}}$$

$$\boxed{V = \frac{kq}{r}}$$

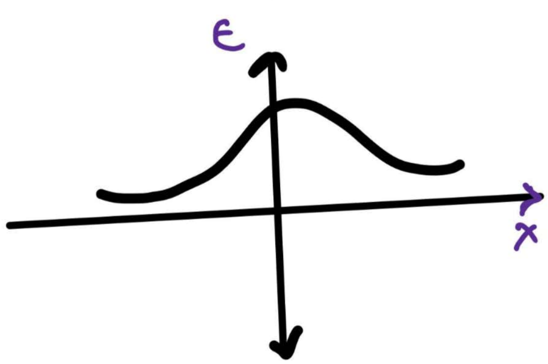
Case 2: Ring



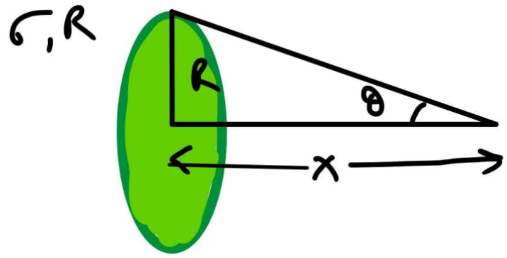


$$\boxed{\vec{E}_P = \frac{kQx}{(R^2+x^2)^{3/2}}}$$

$$\boxed{\vec{V}_P = \frac{kQ}{\sqrt{R^2+x^2}}}$$



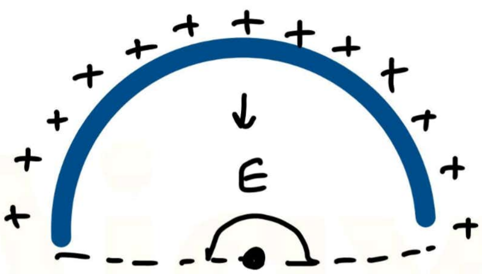
Case 3: Disc



$$\boxed{\vec{E}_P = \frac{\sigma}{2\varepsilon_0}(1-\cos\theta)}$$

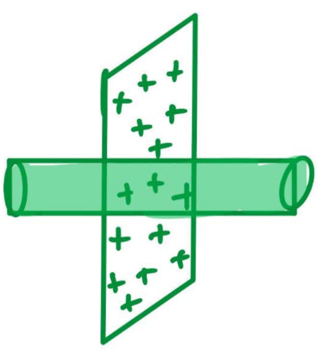
$$\boxed{\vec{V}_P = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{x^2+R^2}+x^2\right)}$$

Case 4: Arc



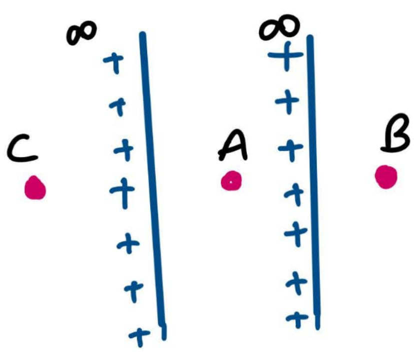
$$\boxed{E_0 = \frac{2k\lambda}{r}\,\frac{\sin\theta}{2}}$$

Case 5: Infinite Sheet



$$\boxed{\vec{E} = \frac{\sigma}{2\varepsilon_0}}$$

$$\boxed{V = V_0 - \frac{\sigma z}{2\varepsilon_0}}$$

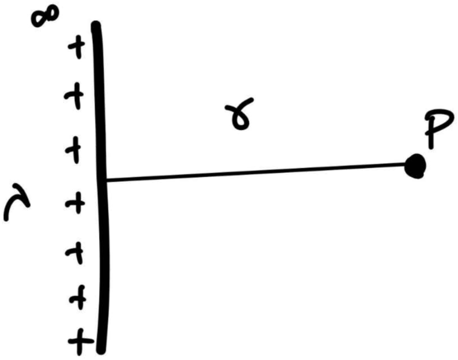


$$\boxed{E_A = -\frac{1}{2\varepsilon_0}(\sigma_1+\sigma_2)}$$

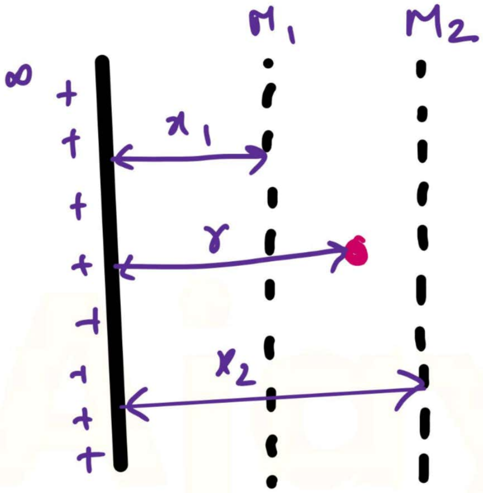
$$\boxed{E_B = \frac{1}{2\varepsilon_0}(\sigma_1-\sigma_2)}$$

$$\boxed{E_C = \frac{1}{2\varepsilon_0}(\sigma_1+\sigma_2)}$$

Case 6: Infinite long wire

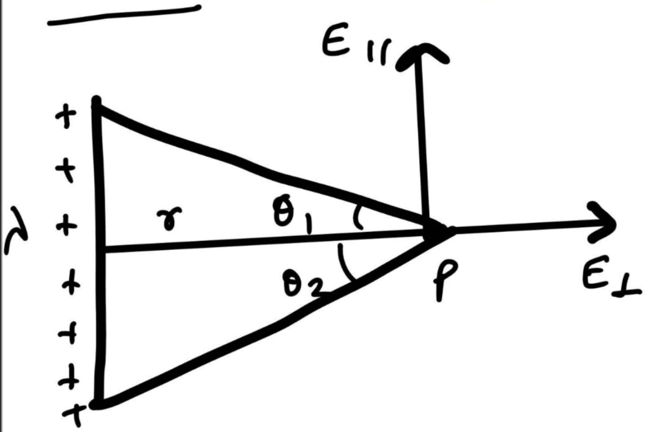


$$\boxed{\vec{E} = \frac{2k\lambda}{r}}$$



$$\boxed{\Delta V = 2k\lambda \ln \frac{x_2}{x_1}}$$

Case 7: Line charge of the finite length of wire.



$$E_{\perp} = \frac{k\lambda}{r}(\sin\theta_1 + \sin\theta_2)$$

$$E_{\parallel} = \frac{k\lambda}{r}(\cos\theta_1 + \cos\theta_2)$$

$$E_{net} = \sqrt{E_{\perp}^2 + E_{\parallel}^2}$$

For semi infinite wire

$\theta_1 = 90^\circ$, $\theta_2 = 0^\circ$

To confirm your JEE Seat

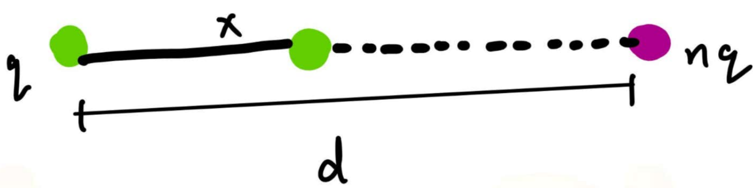
SCAN ME

Case 8: V on a liquid droplets of radius R.

when n small droplets of radius R combine

$$\boxed{V_s = \frac{kQ}{r}} \qquad \boxed{V_b = n^{2/3} V_s} \qquad \boxed{C_b = n^{1/3} C_s}$$

Null point:



• If charge of same sign
  - null point between charges

$$\boxed{x = \frac{d}{\sqrt{n}+1}}$$

• If charges are of opposite sign
  - null point outside charges

$$\boxed{x = \frac{d}{\sqrt{n}-1}}$$

* Null point is always towards the smaller charge.

UNIT 2

CURRENT ELECTRICITY

1) Electric Current $\quad I = \frac{\Delta q}{\Delta t}$

2) Current Density $\quad J = \frac{I}{A}$

3) Drift velocity $\quad V_d = \frac{-e\vec{E}\tau}{m}$

4) Relation b/w current & Drift velocity

$$I = nAeV_d$$

5) Ohm's Law $\quad V = IR$

6) Microscopic form of Ohm's Law

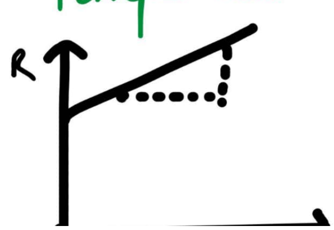
$\boxed{E = \rho J}$ $\quad \boxed{R = \frac{m}{ne^2\tau} \times \frac{l}{A}}$ $\quad \boxed{\rho = \frac{m}{ne^2\tau}}$

7) Mobility of Electron

$\mu = \frac{V_d}{E}$, $\mu = \frac{e\tau}{m}$, $\rho = \frac{1}{ne\mu}$, $\sigma = ne\mu$

8) Temperature Dependence

Metallic conductors $\rightarrow$ T↑ collision↑ τ↓ R↑



$$\rho_t = \rho_0(1 + \alpha t)$$

$$\alpha = \frac{R_2 - R_1}{R_0(t_2 - t_1)}$$

Combination of Resistors

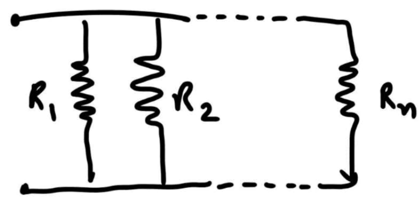
In series

$R_s = R_1 + R_2$

$V_1 = \frac{R_1 V}{R_1 + R_2}$, $V_2 = \frac{R_2 V}{R_1 + R_2}$

Each will have volt $= \frac{V}{n}$

In parallel

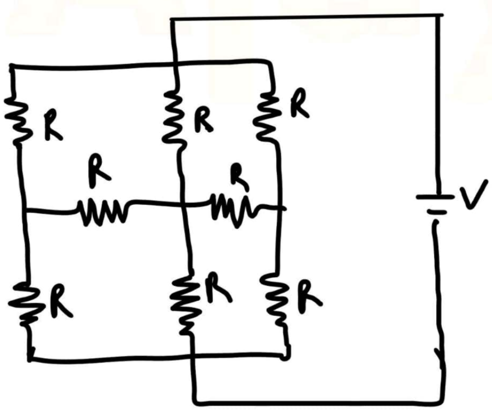


$R_p = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_p = \frac{R_1 R_2}{R_1 + R_2}$

$I_1 = \frac{R_2 I}{R_1 + R_2}$, $I_2 = \frac{R_2 I}{R_1 + R_2}$

I through each $= \frac{I}{n}$

Symmetry Circuit



Rule1: Find plane of symmetry

Rule2: Overlap the circuit about plane of symmetry.

Rule3: Overlapping R will be in parallel

Rule4: Draw new circuit with equivalent resistance after overlapping.

Rule5: If POS ⊥ AB then incoming current is reflected in its corresponding branch, i.e, $I = 0$

Cube

Between two nearer corner = $R_c = \frac{12}{7}R$

Across face diagonal = $R_{fd} = \frac{3}{4}R$

Across main diagonal = $R_{md} = \frac{5}{6}R$

Emf of cell = $\boxed{e = \frac{W}{q}}$

Relation between emf and internal resistance

$$\boxed{e = V + Ir}$$

Combination of cells

Series combination

$$e_{eq} = e_1 + e_2 + \cdots + e_n = ne$$

$$r_{eq} = r_1 + r_2 + \cdots + r_n = nr$$

$$I = \frac{e_{eq}}{(R + r_{eq})} = \frac{ne}{R + nr}$$

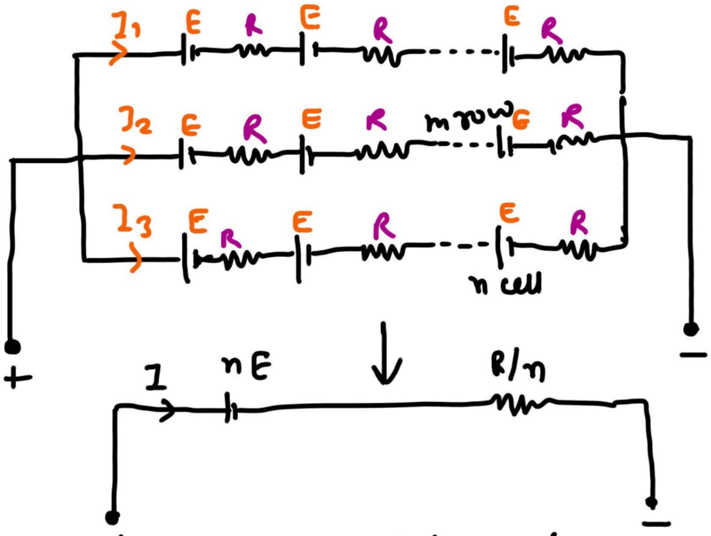
Parallel Combination

$$\frac{1}{r_{eq}} = \frac{1}{r_1} + \frac{1}{r_2} + \cdots + \frac{1}{r_n} = \frac{n}{r}$$

$$r_{eq} = \frac{r}{n}$$

$$I = \frac{e}{\left(R + \frac{r}{n}\right)}$$

Mixed Combination



Series - parallel grouping of cells

$$I = \frac{nE}{\frac{nr}{m} + R}$$

At I_{max}

$$r_{int} = R_{ex}$$

$$\frac{nr}{m} = R$$

Conversion of Galvanometer into

1) Ammeter

(connect low resistance in parallel)

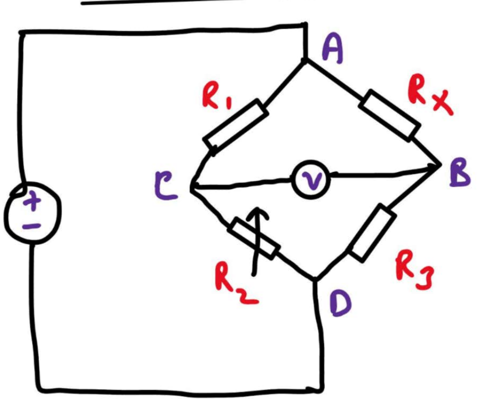
$$R_S = \frac{R_g i_g}{i - i_g}$$

2) Voltmeter

(connect high resistance in series)

$$R = \frac{V}{i_g} - R_g$$

Wheat Stone Bridge



Bridge is balanced when $I_g = 0$

★★ $\boxed{\frac{R_1}{R_2} = \frac{R_x}{R_3}}$

$\frac{R_1}{R_x} > \frac{R_2}{R_3}$ ($I_g \Rightarrow A \longrightarrow D$)

$\frac{R_1}{R_x} < \frac{R_2}{R_3}$ ($I_g \Rightarrow D \longrightarrow A$)

Imp

Kirchoff's Law

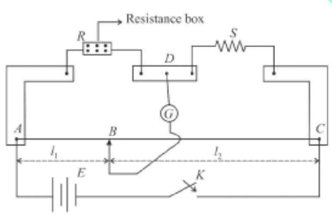
1) **Junction Rule** - The alzebraic sum of all currents meeting at a junction in a closed circuit is zero, i.e,

$$\Sigma I = 0 \quad \text{(at junction)}$$

2) **Loop Rule** - The algebraic sum of all the potential differences in a closed circuit is zero, i.e,

$$\Sigma V = 0$$

Meter Bridge

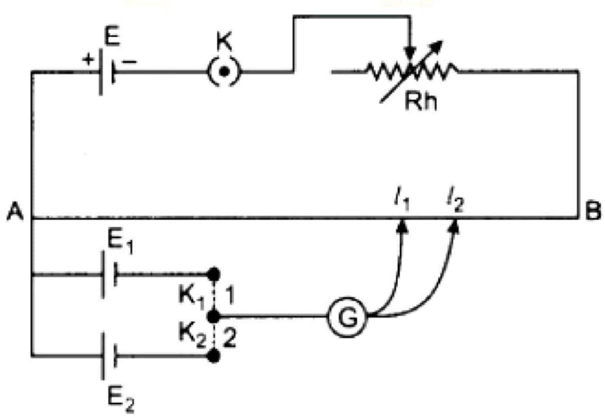


$$\frac{R}{S} = \frac{r l_1}{(100 - l_1)}$$

$$R = \frac{l_1 . S}{100 - l_1}$$

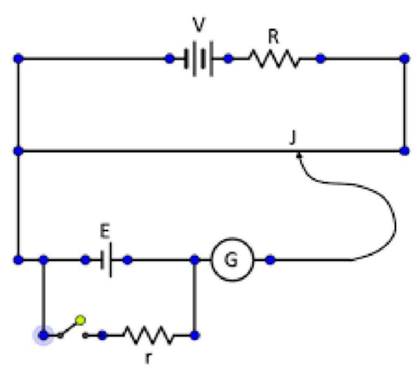
Potentiometer

Comparison of EMF of two cells



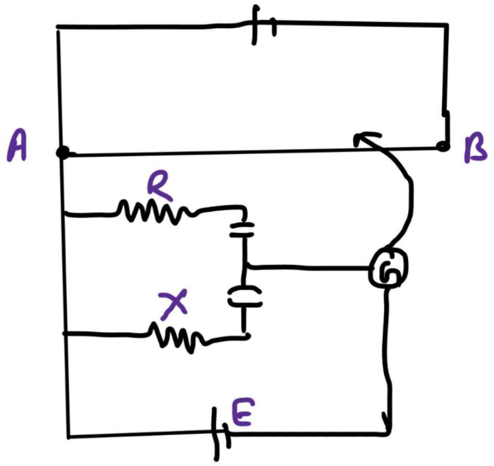
$$\boxed{\frac{E_1}{E_2} = \frac{l_1}{l_2}}$$

Measurement of Internal resistance of a battery



$$\boxed{r = R = \frac{R(l_1 - l_2)}{l_2}}$$

Comparison of two resistance



$$\frac{R_1+R_2}{R_1} = \frac{l_2}{l_1}$$

$$\frac{R_2}{R_1} = \frac{l_2-l_1}{l_1}$$

Measurement of current = $I = \frac{l_2}{R} \times \frac{E_1}{l_1}$

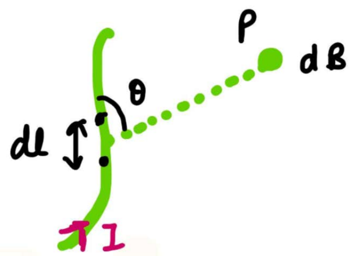
UNIT 3

MOVING CHARGES

Biot Savart Law

$$\vec{dB} = \frac{\mu_0}{4\pi}\left(\frac{\vec{dl}\times\vec{r}}{r^3}\right)$$

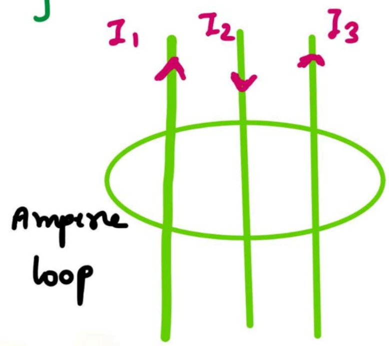
Direction $\vec{B} = \vec{dl}\times\vec{r}$



Ampere Circuital Law

$$\int \vec{B}.\vec{dl} = \mu_0(I_i)$$

$$\int \vec{B}.\vec{dl} = \mu_0(I_1 + I_3 - I_2)$$



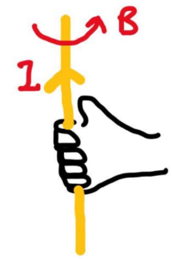
Direction

Right Hand Thumb Rule

Curl Finger — $\vec{B}$

Thumb — Current

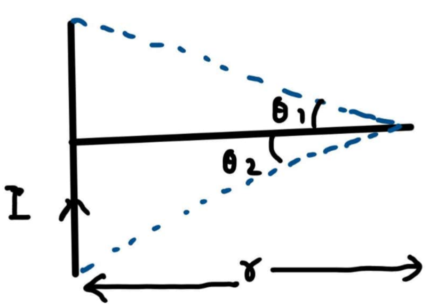
Second method



Current

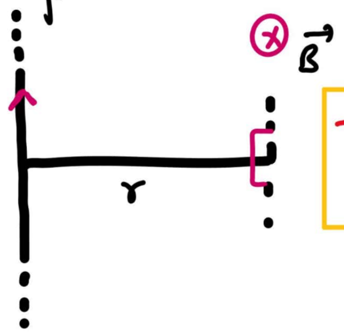
$\vec{B}$

I) Straight current carrying conductor:



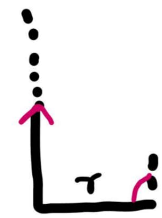
$$\boxed{B = \frac{\mu_0 I}{4\pi r}(\sin\theta_1 + \sin\theta_2)}$$

II) Infinite wire current carrying conductor:



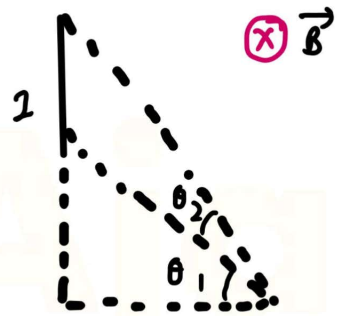
$$\vec{B} = \frac{\mu_0 I}{2\pi r}$$

III) Semi Infinite wire



$$\vec{B} = \frac{\mu_0 I}{4\pi r}$$

IV) Infinite wire current carrying conductor

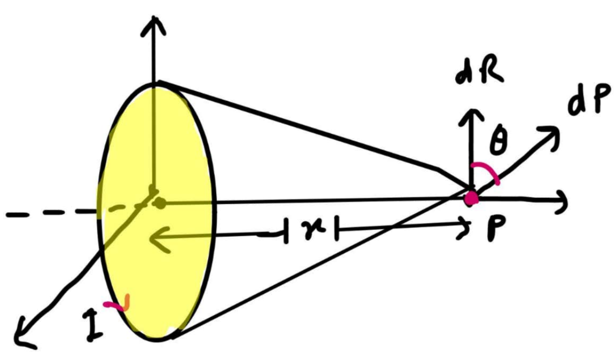


$$B = \frac{\mu_0 I}{4\pi r}(\sin\theta_2 - \sin\theta_1)$$

V) Circular Arc

$$|B| = \frac{\mu_0 I}{2r}\left(\frac{\theta}{360^\circ}\right)$$

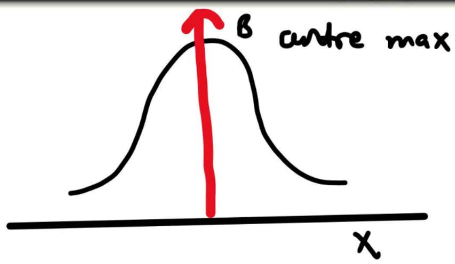
VI) Magnetic field on axis of current carrying coil



$$B = \frac{\mu_0 I R^2}{2(x^2+R^2)^{3/2}}\hat{i}$$

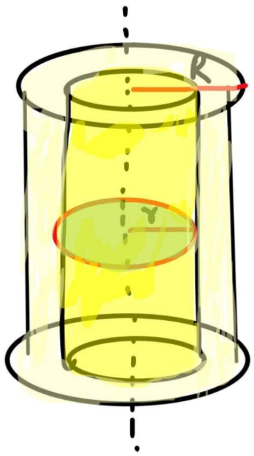
if coil has n turns

$$B = \frac{\mu_0 n I R^2}{2(x^2+R^2)^{3/2}}\hat{i}$$



$$\boxed{B_{max} = \frac{\mu_0 n I}{2R}} \text{ at } x = 0$$

VII) Hollow Cylinder current carrying conductor:



$r < R$ (internal)

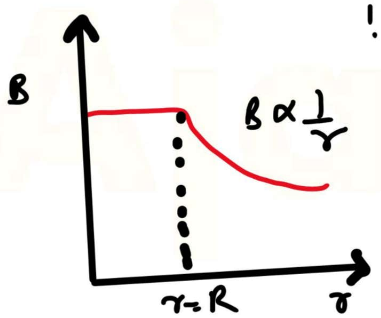
$\oint \vec{B}.\vec{dl} = \mu(0)$ $\boxed{B_i = 0}$

$r = R$ (surface)

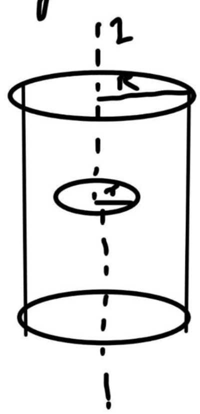
$$\boxed{B_s = \frac{\mu_0 I}{2\pi R}}$$

$r > R$ (external)

$\oint \vec{B}.\vec{dl} = \mu_0 I$ $\boxed{B = \frac{\mu_0 I}{2\pi r}}$



VIII) Solid Cylinder current carrying conductor:



$r < R$ (internal)

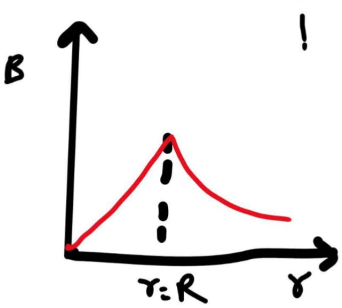
$$\boxed{B = \frac{\mu_0 I}{2\pi R^2}}$$

$r = R$ (surface)

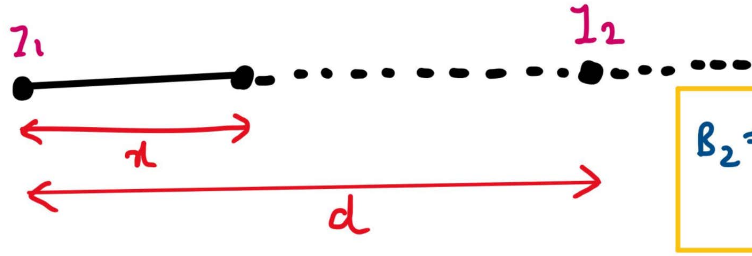
$$\boxed{B = \frac{\mu_0 I}{2\pi R}}$$

$r > R$ (external)

$$\boxed{B = \frac{\mu_0 I}{2\pi r}}$$



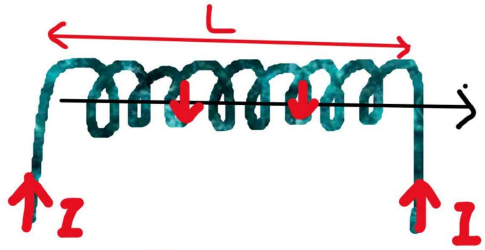
Null Point



$$\boxed{B_1 = \frac{\mu_0 I_1}{2\pi x}}$$

$$\boxed{B_2 = \frac{\mu_0 I_1}{2\pi(d-x)}}$$

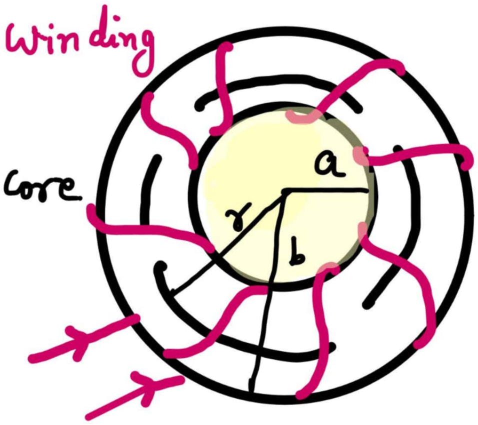
IX) Solenoid



$$\boxed{B_{centre} = \mu_0 n I}$$

$$\boxed{n = \frac{N}{L}}$$

X) Torroid



$r < a \qquad B = 0$

$r > b \qquad B = 0$

$a < r < b \qquad B = \frac{\mu_0 N I}{2\pi r}$

Magnetic Dipole Moment

$$\vec{P_m} = IA\hat{n}$$

If loop carries N no. of loops

$$\vec{P_m} = NIA$$

Magnetic field / Torque / Work Done / PE due to dipole

$\vec{B}$ axial due to dipole at a distance r from centre

$$E_1 = \left(\frac{\mu_0}{4\pi}\right)\frac{2p_m}{r^3}$$

$\vec{B}$ equitorial due to dipole at a distance r from centre

$$E_\perp = \left(\frac{\mu_0}{4\pi}\right)\frac{p_m}{d^3}$$

Torque if placed in an external field

$$\vec{\tau} = \vec{p_m} \times \vec{E}$$

work done in rotating the dipole

$$W = p_m B(1-\cos\theta)$$
$$= p_m B - \vec{p_m}\,\vec{B}$$

P.E. in any external field

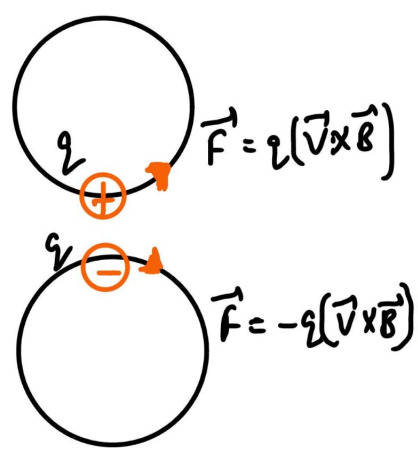
$$U = -\vec{p_m}\,\vec{B}$$

Motion of the charged particle in B

Force on charge particle = $\boxed{\vec{F} = q(\vec{V} \times \vec{B})}$

work done = 0

$\therefore$ F is $\perp$ to displacement



$$r = \frac{mV}{qB} = \frac{p}{qB} = \frac{\sqrt{2mKE}}{qB}$$

$$\omega = \frac{V}{r} = \frac{qB}{m}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

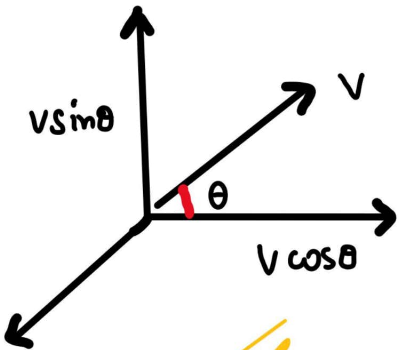
To get the best Question Practice material

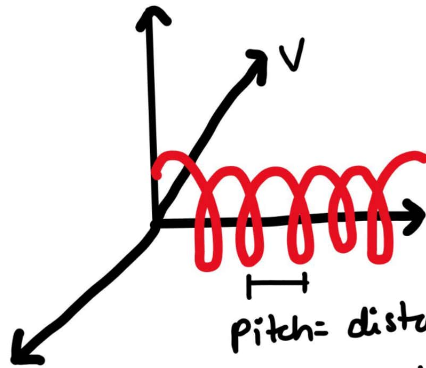
SCAN ME

when charge is projected at any angle θ

Trajectory → Helical

$$\text{pitch} = \frac{v\cos\theta \, 2\pi m}{qB}$$





pitch = distance drift in 1 rotation

Lorentz force

$$\vec{F}_{total} = q(\vec{E}) + q(\vec{v}\times\vec{B})$$

Cyclotron

$$\boxed{r = \frac{mv}{qB}}$$

v↑ r↑

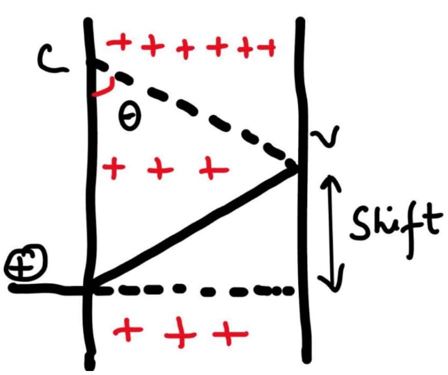
$$\boxed{V_{max} = \frac{qBR}{m}}$$

$$\boxed{KE = \frac{q^2B^2R^2}{2m}}$$

$$\boxed{T = \frac{2\pi m}{B}}$$

$$\boxed{\nu = \frac{1}{T} = \frac{qB}{2\pi m}}$$

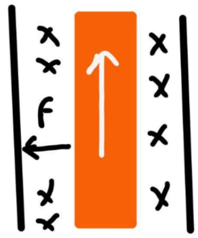
Angle of deflection if a charge passes through B :



$$\boxed{\sin\theta = \frac{dqB}{mv}}$$

$$\boxed{\text{Shift} = \frac{mv}{qB}\left(1-\sqrt{1-\sin^2\theta}\right)}$$

force on current carrying conductor



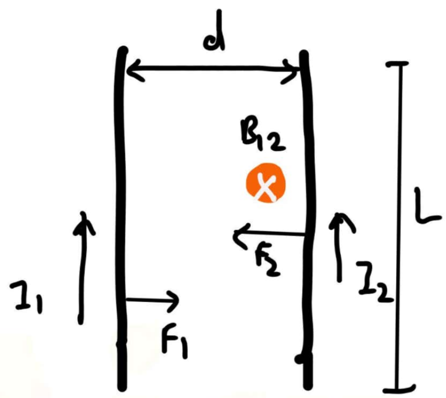
$$\boxed{\vec{F} = I(\vec{L} \times \vec{B})}$$

$|F| = BIL\sin\theta$

if $\theta = 90°$

$|F| = BIL$

force between two parallel wires

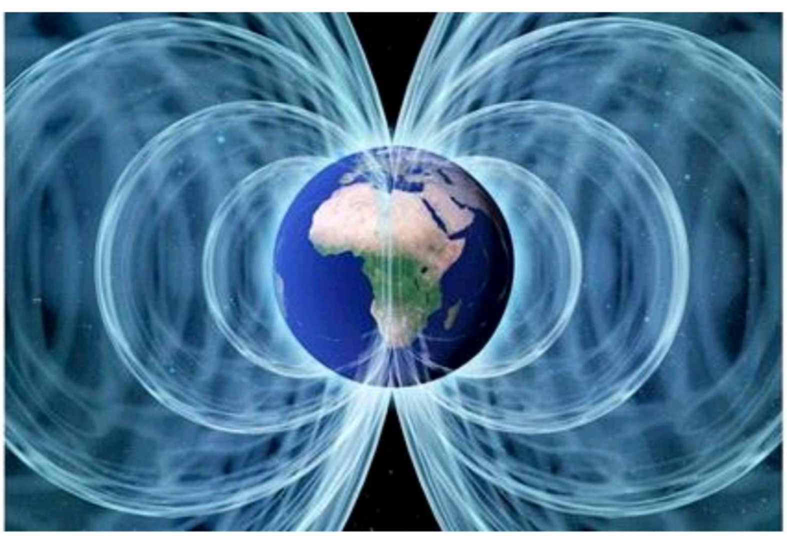


$$\boxed{\frac{F}{L} = \frac{\mu_0 2 I_1 I_2}{4\pi d}}$$

Direction of I

Same: Attractive force

Opp: Repulsive force

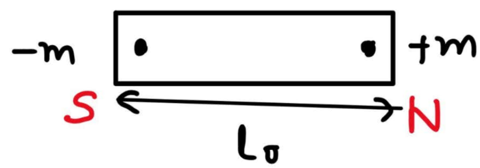


MAGNETISM AND MATTER

Magnetic Dipole Moment (Bar Magnet)

$\vec{M} = m\vec{l}$

$l \cong 0.91 l_0$



If two bars magnets of equal dipole moment are placed at same angle.

At $\theta = 0°$ $\quad \vec{M}_{net} = 2M$

At $\theta = 180°$ $\quad \vec{M}_{net} = 0$

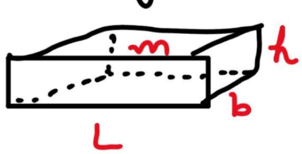
At $\theta = 90°$ $\quad \vec{M}_{net} = \sqrt{2}M$

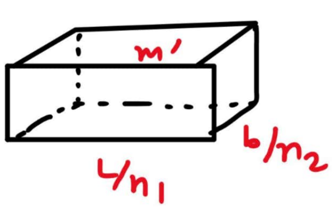
At $\theta = 120°$ $\quad \vec{M}_{net} = M$

Closed Polygon $\quad \vec{M}_{net} = 0$

Square $\quad \vec{M}_{net} = 2\sqrt{2}M$

Cutting of Magnets :





$n_1 \Rightarrow$ no. of part lengths is broken into

$n_2 \Rightarrow$ no. of parts breadth is broken into

$$\vec{M}_{net} = \frac{M_i}{n_2 n_1}$$

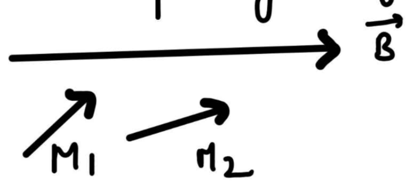
If bar magnet is cut into two equal parts:

Parallel to its breadth.

$\perp$ to its length

$$\vec{M}_{net} = \frac{Mi}{2}$$

Comparing magnetic moments of magnets of same size:



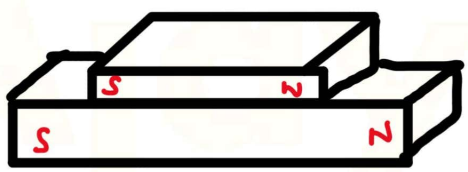
oscillating Dipole

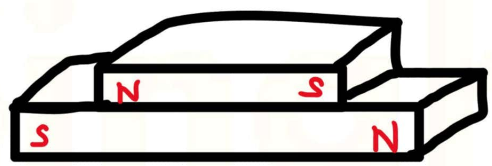
$$T_1 = 2\pi\sqrt{\frac{I}{M_1 B_H}} \qquad T_2 = 2\pi\sqrt{\frac{I}{M_2 B_H}}$$

where I = Moment of Inertia

$$\frac{M_1}{M_2} = \frac{T_2^2}{T_1^2}$$

Comparing magnetic moments of magnets of different sizes:





$$T_1 = 2\pi\sqrt{\frac{I_1 + I_2}{(M_1 + M_2) B_H}}$$

$$T_1 = 2\pi\sqrt{\frac{(I_1 + I_2)}{(M_1 - M_2) B_H}}$$

Magnetic Flux

for uniform B

$$\phi = \vec{B}\cdot\vec{A} = BA\cos\theta$$

for variable B

$$\int d\phi = \int B \cdot dA$$

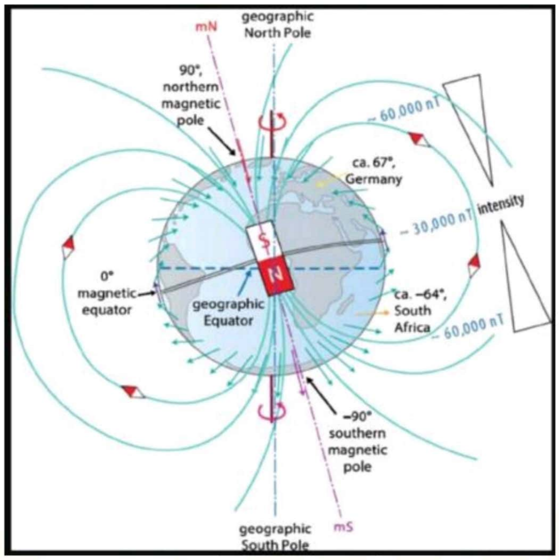
$$\phi = \int B \cdot dA$$

Gauss's Law

The net magnetic flux through a closed surface S is zero.

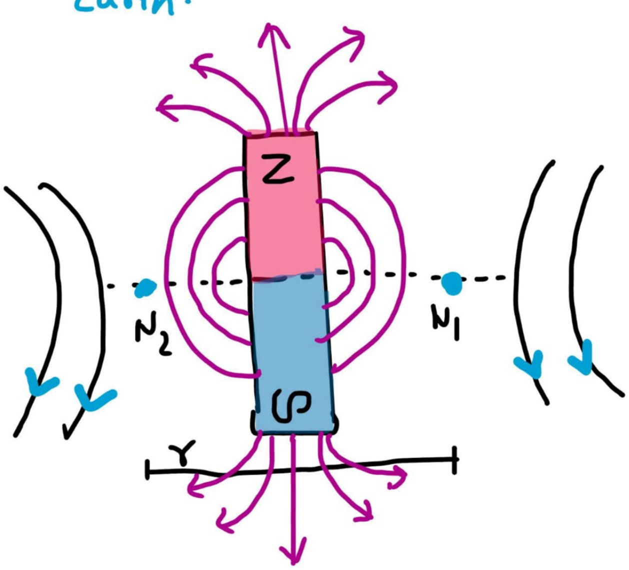
$$\phi_B = \oint_S \vec{B}\cdot\vec{dS} = 0$$

Geo Magnetism



Neutral point

If north pole of bar magnet points towards north of Earth.

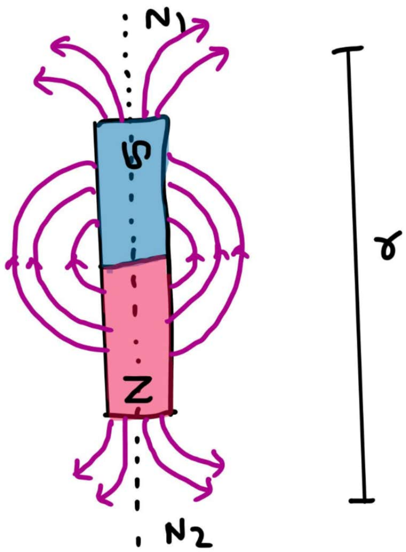


$$B_H = B_e = \frac{\mu_0}{4\pi}\frac{m}{(r^2+l^2)^{3/2}}$$

if $l <<< r$

$$B_H = B_e = \frac{\mu_0 m}{4\pi r^3}$$

If South pole of Bar Magnet points towards North of Earth:



$$B_H = B_a = \frac{\mu_0}{4\pi} \frac{2mr}{(r^2 - l^2)^2}$$

$$m = \frac{4\pi r^3 B\pi}{2\mu_0}$$

Magnetic Induction

$B = B_0 + B_m$ $\qquad$ $B = \mu_0(NI + M)$

$B_0 =$ Magnetic field due to solenoid

$B_m =$ Magnetic field due to magnetised substance

$B =$ Magnetic Induction

1) Magnetising Field or Magnetising Intensity

$$\vec{H} = \frac{\vec{B_0}}{\mu_0} \qquad \vec{H} = NI$$

2) Intensity of Magnetisation

$$I = \frac{m_{net}}{V} \qquad I = \frac{m}{A}$$

3) Magnetic permeability

$$\mu = \frac{B}{H}$$

4) Relative magnetic permeability

$$\mu_r = \frac{\mu}{\mu_0} \quad , \quad \mu_r = \frac{B_m}{B_0}$$

5) Magnetic susceptibility

$$\chi_m = \frac{I}{H}$$

* Relation between μ_r & χ_m

$$\boxed{\mu_r = 1 + \chi_m}$$

Atomic Magnetism

Radius of any orbit	$r_n = \frac{a_0 n^2}{Z}$	$r \propto \frac{1}{Z}$
Velocity of any orbit	$v_n = v_0 Z/n$	$v \propto Z$
Time period of electron In any orbit	$T_n = \frac{2\pi r_n}{v_n}$	$T \propto \frac{r^3}{Z^2}$
Frequency of electron	$\nu = 1/T$	$\nu \propto \frac{Z^2}{n^3}$
Current in any orbit	$I = q/t$	$I \propto \frac{Z^2}{n^3}$
Magnetic field in any orbit	$B = \frac{\mu_0 I}{2r_n}$	$B \propto Z^3/n^5$
Magnetic moment in any orbit	$M = IA$	$M \propto n$
Magnetic moment linked with Angular momentum	$\frac{\lvert\vec{M}\rvert}{\lvert\vec{L}\rvert} = \frac{e}{2M}$	

Tangent Galvanometer

$$B = \frac{\mu_0 I N}{2r}$$

I: current through coil

N = No. of turn

r: radius of coil

$$I = \frac{2r B_H}{\mu_0 N}\tan\theta \quad , \quad I = k\tan\theta$$

Vibration Magnetometer

compares intensity of earth's B at two places

$$T = 2\pi\sqrt{\frac{I}{MH}}$$

T : Time period

I = Moment of Inertia

M : Magnetic Moment

H = Magnetic Field Intensity

$$\frac{H_2}{H_1} = \frac{T_1^2}{T_2^2}$$

Moving coil Galvanometer

$I \propto \theta$ $\quad\theta = GI$

$$G = \frac{nBA}{K}$$

Ferromagnets

soft	Hard
low retentivity	High retentivity
low coerivity	High coercivity
Small Hysteresis	Large Hysteresis
low area	Large area
↓ loss per cycle	↑ loss per cycle

ELECTROMAGNETIC INDUCTION

Magnetic flux

$\phi = \vec{B} \cdot \vec{A}$

$\phi = NBA\cos\theta$

if B is varying

$\int \phi = \int B \cdot dA$

Flux through a loop placed $\perp$ to a current carrying wire

$\phi_{loop} = 0$

flux through ring where $l >> r$

$$\phi = \frac{2\sqrt{2}\mu_0 I R^2}{l}$$

Faraday's Law

(i) whenever the magnetic flux linked with a circuit changes, an induced emf is produced in it.

(ii) The induced emf lasts, so long as the change in magnetic flux continues

The magnitude of induced flux is directly proportional to the rate of change in mag. flux, i.e,

$$\varepsilon = -N\frac{d\phi}{dt} \qquad I = -\frac{\varepsilon}{R} = \frac{d\phi/dt}{R}$$

Lenz's Law

The polarity of induced emf is s.t it tends to produce a current which opposes the change in ϕ that produced it.

coil is in Inertia of flux.

$\phi \odot \uparrow \longrightarrow B \otimes$

$\phi \odot \downarrow \longrightarrow B \odot$

$\phi \otimes \uparrow \longrightarrow B \odot$

$\phi \otimes \downarrow \longrightarrow B \otimes$

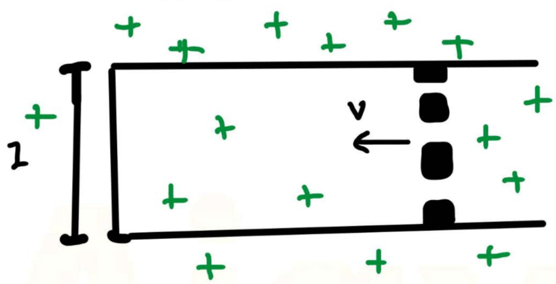
ϕ = constant

$I = 0$

$\odot$ outward $\vec{B}$

$\otimes$ Inward $\vec{B}$

Motional emf



$$\boxed{\varepsilon = Blv}$$

$$\boxed{I = \frac{Blv}{R}}$$

Eddy Current

Induced circulating currents produced in a metallic block itself due to change in B.

$$\boxed{I = -\frac{\varepsilon}{R} = \frac{d\phi/dt}{R}}$$

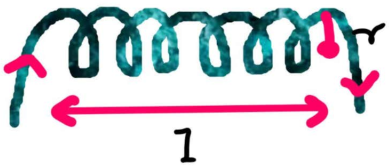
Rotational EMF $\quad \varepsilon = \frac{1}{2}B\omega l^2$

Dissipated power in a conductor moving in a B $\quad P = \frac{B^2 v^2 l^2}{R}$

Induced electric field due to time varying B $\quad \left|\oint \vec{E}\cdot\vec{dl}\right| = \left|-\frac{d\phi}{dt}\right|$

Self Inductance

(Inertia of electricity)



$\boxed{L = \mu_0 n^2 A l}$

$\boxed{\phi = L I}$

Energy stored in an inductor

$W = \frac{1}{2} L I_0^2 \quad , \quad U_m = \frac{1}{2} \frac{B^2 A l}{\mu_0}$

Magnetic Energy Density per unit volume

$\frac{1}{2} \mu_0 B^2$

Electric energy density per unit volume

$\frac{\varepsilon_0 E^2}{2}$

I = constant

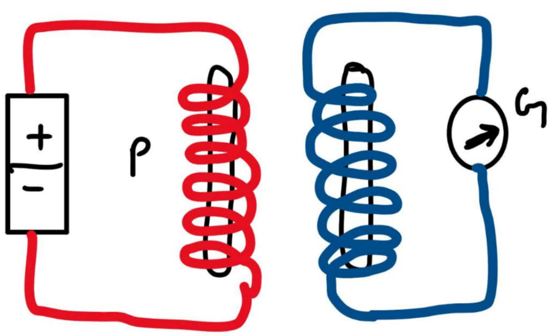
E = 0

I(↑)

E opp. to I

I(↓) E along I

Mutual Inductance



$|\varepsilon| = M \frac{dI}{dt}$

$\phi_s = M I_p$

To take your preparation to next level by testing

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SCAN ME

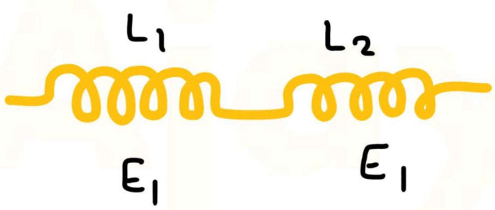
For coaxial solenoid of equal length $\quad M = \mu_0 n_1 n_2 A l$

Two concentric coils of diff. radii $\quad M = \left(\frac{\mu_0}{4\pi}\right) \frac{2\pi^2 r_1^2}{r_2}$

Two coaxial circular coil of diff. radii $\quad M = \left(\frac{\mu_0}{4\pi}\right) \frac{2\pi^2 R^2 r^2}{(R^2+x^2)^{3/2}}$

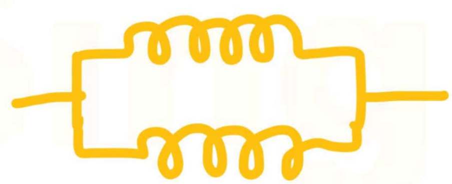
Combination

Series



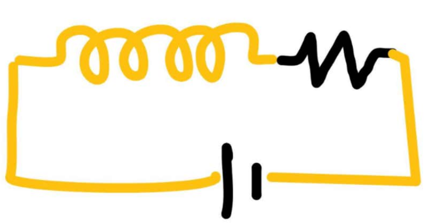
$$\boxed{L_{eq} = L_1 + L_2}$$

Parallel



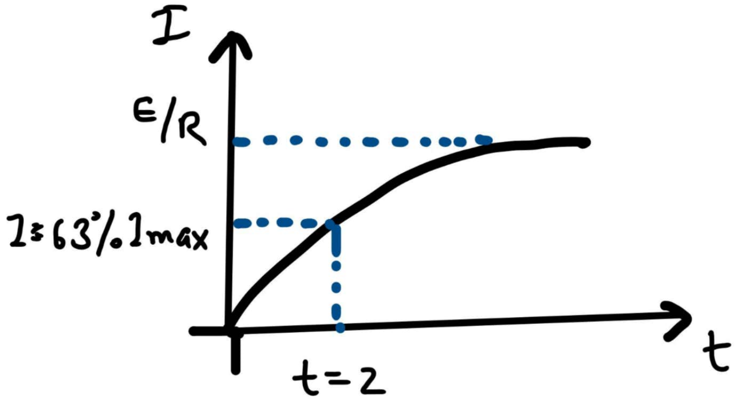
$$\boxed{L_{eq} = \frac{L_1 L_2}{L_1 + L_2}}$$

RL Circuit



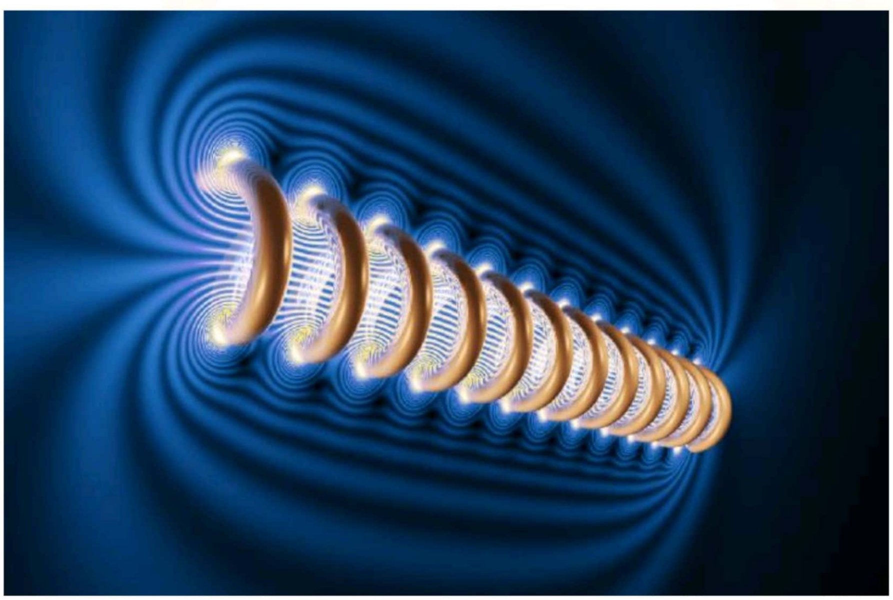
$$\boxed{I = I_0\left(1 - e^{-tR/L}\right)}$$

$$I_{max} = \frac{E}{R}$$



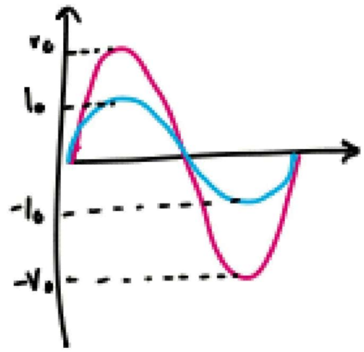
Decay of current in the RL circuit :

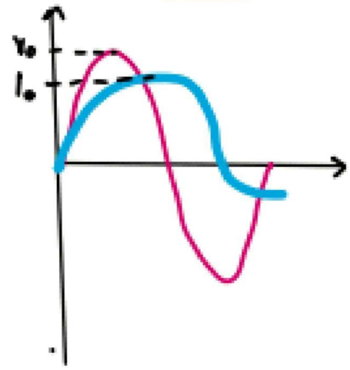
$$\boxed{I_{max} = \frac{E}{R}} \qquad \boxed{I = I_0 e^{-tR/L}}$$

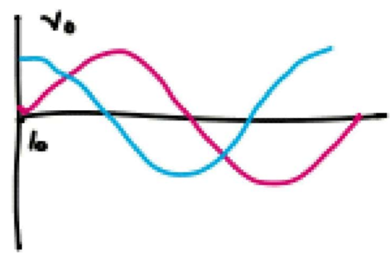


	R	L	C

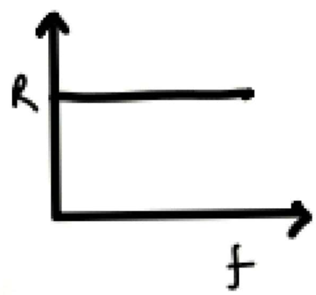
Circuit

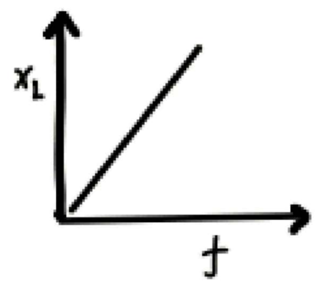


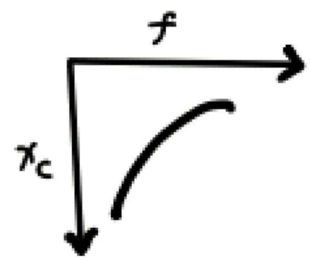




Variation Z with f



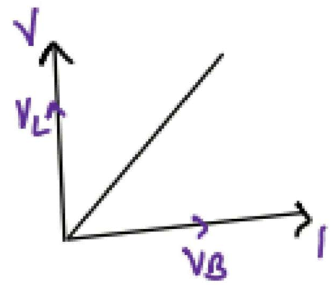




CIRCUIT COMBINATION

R-L — Circuit

Phasor



Phase Difference

V leads I

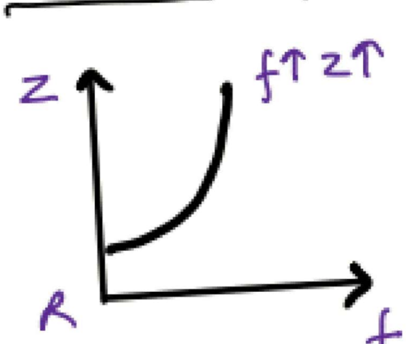
$\phi = 0$ to $\frac{\pi}{2}$

Impedence

$Z = \sqrt{R^2 + X_L^2}$

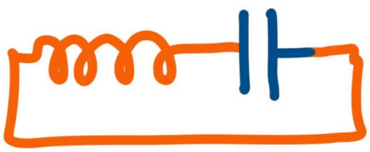
$= \sqrt{R^2 + \omega^2 L^2}$

Variation (Z/f)

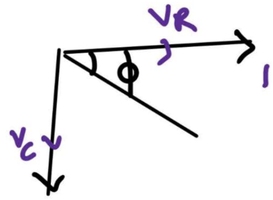


R–C

Circuit



Phasor



Phase Difference

V lags I

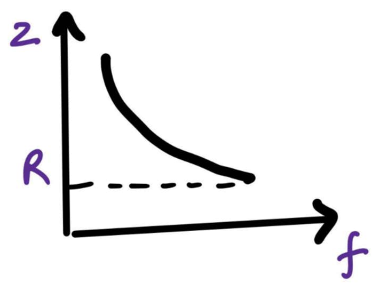
$\phi = -\frac{\pi}{2}$ to 0

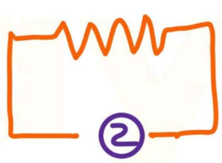
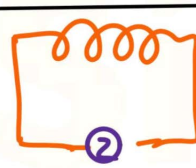
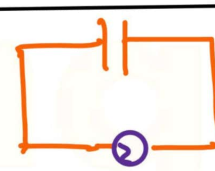
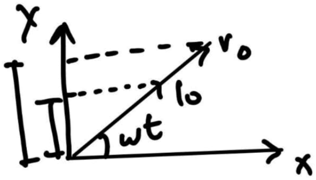
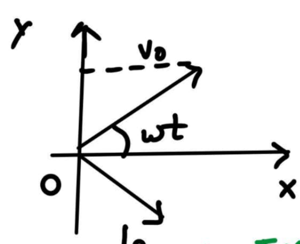
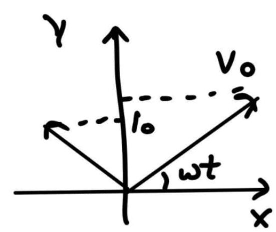
Impedence

$$Z = \sqrt{R^2 + X_C^2}$$

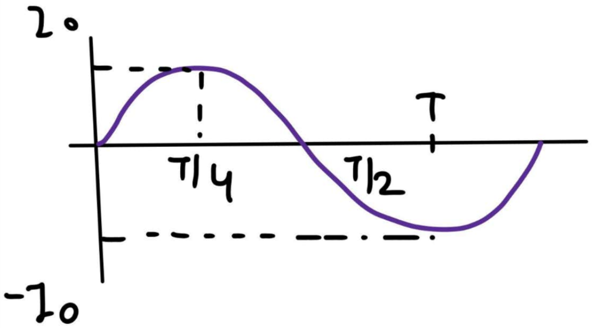
$$= \sqrt{R^2 + \left(\frac{1}{\omega C}\right)^2}$$

Variation (Z|f)



	R	L	C
Circuit			
Supply voltage	$V = V_0 \sin\omega t$	$V = V_0 \sin\omega t$	$V = V_0 \sin\omega t$
Current	$I = I_0 \sin\omega t$	$I = I_0 \sin(\omega t - \frac{\pi}{2})$	$I = I_0 \sin(\omega t + \frac{\pi}{2})$
Comparison	V & I are in same phase	I lags by $\frac{\pi}{2}$	I leads by $\frac{\pi}{2}$
Peak current	$I_0 = \frac{V_0}{R}$	$I_0 = \frac{V_0}{\omega L}$	$I_0 = V_0 \omega C$
Impedence (Z)	$R = \frac{V_0}{I_0}$	$X_L = \omega L$	$X_C = \frac{1}{\omega C}$
Phasor			
Ohm's law	$V = IR$	$V_L = I X_L$	$V_C = I X_C$
Power	$P_{av} = \frac{1}{2} I_0^2 R$	$P_{av} = 0$	$P_{av} = 0$

ALTERNATING CURRENT



$I = I_0 \sin \omega t$

$\omega = \frac{2\pi}{T} = 2\pi f$

Alternating Voltage:

$\boxed{V = V_0 \sin \omega t + \theta}$

$\boxed{V_0 = NBA\omega}$

Average or Mean Value

Current

$$I_{av} = \frac{\int_{t_1}^{t_2} I(t)\,dt}{t^2 - t_1}$$

Voltage

$$V_{av} = \frac{\int_{t_1}^{t_2} V(t)\,dt}{t_2 - t_1}$$

$I_{av} = \frac{2I_0}{\pi} = 63.7\%$ of I_0

$V_{av} = \frac{2}{\pi} V_0 = 63.7\%$ of V_0

Root Mean Square Value

Current

$$I_{rms} = \frac{\int_{t_1}^{t_2} I^2(t)\,dt}{t_2 - t_1}$$

Voltage

$$V_{rms} = \frac{\int_{t_1}^{t_2} V^2(t)\,dt}{t_2 - t_1}$$

$I_{rms} = \frac{I_0}{\sqrt{2}} = 70.7\%$ of I_0

$V_{av} = \frac{2V_0}{\pi}$ 63.7% of V_0

Some important values :

$I = a\sin\theta$ $\qquad I_{max} = a$

$I = a + b\sin\theta$ $\qquad I_{max} = a + b$

$I = a\sin\theta + b\cos\theta$ $\qquad I_{max} = \sqrt{a^2 + b^2}$

$I = a\sin^2\theta$ $\qquad I_{max} = a$

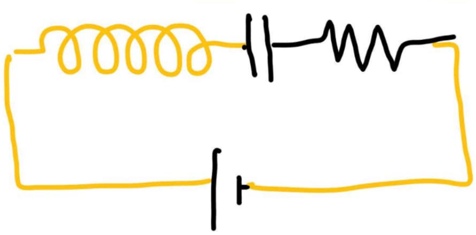
$$\langle \sin\theta \rangle = \langle \sin 2\theta \rangle = 0$$

$$\langle \cos\theta \rangle = \langle \cos 2\theta \rangle = 0$$

$$\langle \sin\theta\cos\theta \rangle = 0$$

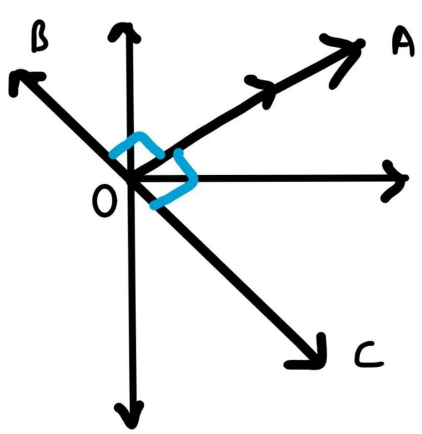
$$\langle \sin^2\theta \rangle = \langle \cos^2\theta \rangle = \frac{1}{2}$$

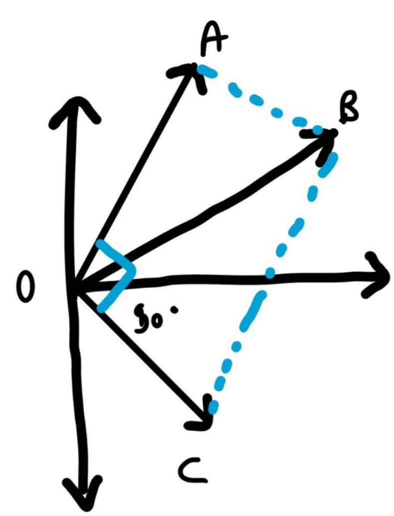
LCR Circuit



$$\boxed{V = V_0 \sin\omega t}$$

$$\boxed{I = I_0(\sin\omega t + \phi)}$$





$V_{net} = V_R^2 + (V_L - V_C)^2$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

$\tan\phi = \dfrac{V_L - V_C}{V_R}$

$= \dfrac{X_L - X_C}{X_R}$

$I_0 = \dfrac{V_0}{\sqrt{R^2 + (X_L - X_C)^2}}$

★ $X_L = X_C$ $\tan\phi = 0$ Pure resistive circuit

★ $X_L > X_C$ $\tan\phi = -ve$ Inductance dominated circuit

★ $X_L < X_C$ $\tan\phi = +ve$ Capacitance dominated circuit

1) Resonance Frequency

$\boxed{\omega_r = \dfrac{1}{\sqrt{LC}}}$ $\boxed{f_r = \dfrac{1}{2\pi\sqrt{LC}}}$

2) Band width & Q-factor

$\boxed{\Delta\omega = \omega_2 - \omega_1}$ $\boxed{Q = \dfrac{\omega_0}{2\Delta\omega} = \dfrac{1}{R}\sqrt{\dfrac{L}{C}}}$

3) Choke coil used to control current in AC circuit

$\boxed{\cos\phi = \dfrac{R}{R^2 + \omega^2 L^2}}$

4) Transformer

$$\boxed{\frac{V_s}{V_p} = \frac{N_s}{N_p}} \qquad \boxed{\frac{I_s}{I_p} = \frac{N_p}{N_s}} \qquad \boxed{\frac{V_s}{V_p} = \frac{I_s}{I_p} = \frac{N_p}{N_s}}$$

Setup Transformer

$V_{output} > V_{input}$

$N_s > N_p$

$I_s < I_p$

$P_{input} = P_{output}$

Set Down Transformer

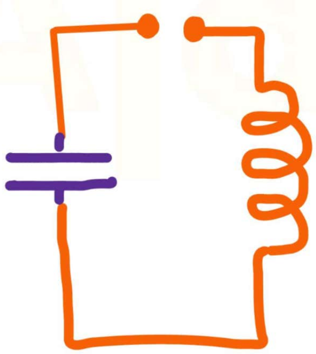
$V_{output} < V_{input}$

$N_s < N_p$

$I_s > I_p$

$P_{input} = P_{output}$

LC Oscillations



$$Q(t) = Q_0 \cos \omega_0 t$$

$$I(t) = I_0 \sin \omega_0 t$$

Energy

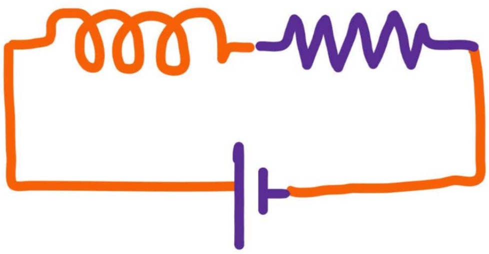
$$U = U_L + U_C$$

$$U_B = \frac{L I_0^2}{2} \sin^2 \omega t$$

$$U = \frac{1}{2} L I^2 + \frac{1}{2} \frac{Q^2}{C}$$

$$U_E = \frac{Q_0^2}{2C} \cos^2 \omega_0 t$$

Damped Oscillations

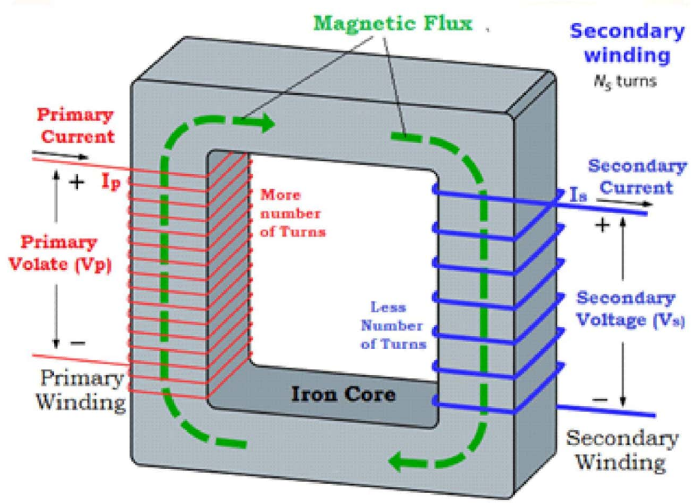


$$\boxed{\frac{d^2q}{dt^2} + \frac{R}{L}\frac{dq}{dt} + \frac{q}{LC} = 0}$$

$$q = Q_{max}\, e^{-bt/2m}$$

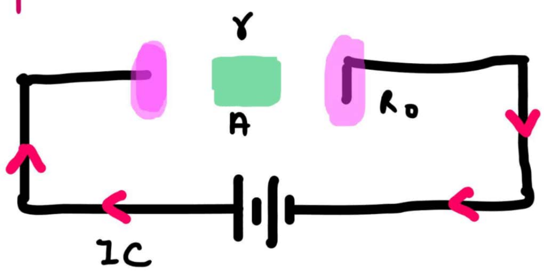
$$q = Q_{max}\, e^{-Rt/2m}$$

$$Q^2 = e^{-kt/2}$$



EM WAVES

1) Displacement Current



Displacement current results due to change in electric field

2) Charge on capacitor at any time t

$$q = CV\left[1 - e^{-t/RC}\right]$$

3) Conduction current

$$I_e = \frac{dq}{dt} = \frac{V}{R}e^{-t/RC}$$

4) Electric field varying with time b/w the plates of capacitor

$$\vec{E}_E = \frac{CVe^{-t/RC}}{\pi R_0^2 \varepsilon_0}$$

5) Flux at A if cylinder is drawn at surface

$$\phi_E = \frac{CVr^2}{\varepsilon_0 R_0^2}\left[1 - e^{-t/RC}\right]$$

6) Charge between plates of capacitor

$$q_E = \varepsilon_0\left[\frac{CVr^2}{\varepsilon_0 R_0^2}\left[1 - e^{-t/RC}\right]\right]$$

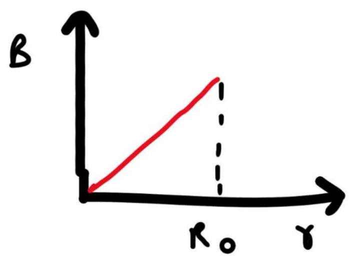
7) Displacement current

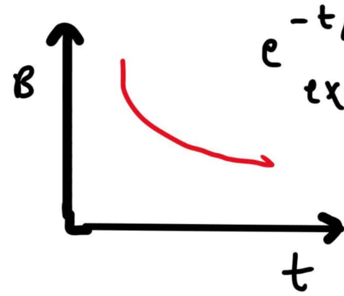
$$I_{dis} = \frac{V}{R}\left[1 - e^{-t/RC}\right]\left[\frac{r^2}{R_0^2}\right]$$

$$I_d = \varepsilon_0\left(\frac{d\phi_E}{dt}\right)$$

8) Magnetic field b/w plates of capacitor

$$\vec{B} = \frac{\mu_0 V}{2\pi R} e^{-t/RC} \frac{r^2}{R_0^2}$$





$e^{-t/RC}$ exponential decay

Important

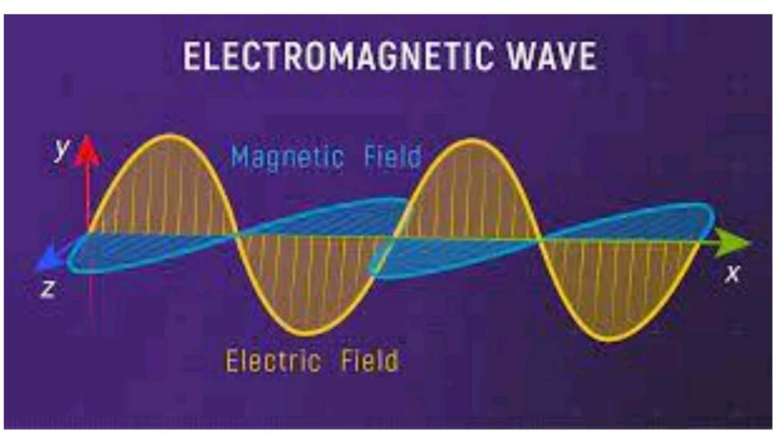
Maxwell's Equations

1) Gauss's law of Electricity $\quad \oint E \cdot dA = \frac{Q}{\varepsilon_0}$

2) Gauss's law of Magnetism $\quad \oint B \cdot dA = 0$

3) Faraday's law $\quad \oint E \cdot dl = -\frac{d\phi_B}{dt}$

4) Ampere - Maxwell's law $\quad \oint B \cdot dl = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d\phi_c}{dt}$



EM waves

$y = A\sin(kx \pm \omega t)$

$2\pi\omega = \frac{2\pi}{T}$

$V_{max} = \frac{\omega}{K}$

y = Displacement of particles

A = Amplitude

x = direction of propagation of wave

ω = Angular frequency

k = wave No $= \frac{2\pi}{\lambda}$

Speed

In vaccuum

$C_{vacuum} = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$

In medium

$C_{medium} = \frac{1}{\sqrt{\mu_m \varepsilon_m}}$

$$V = \frac{1}{\sqrt{\varepsilon_r \varepsilon_0 \mu_r \mu_0}}$$

Refractive Index

$\frac{C}{V} = \sqrt{\varepsilon_r \mu_r}$

Poynting vector

$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B})$

Intensity

$\vec{I} = \frac{\text{Power}}{\text{Area of wavefront}}$

$\vec{I} = \frac{1}{2}\varepsilon_0 E_0^2 C$

$\vec{I} = \frac{B_0^2}{2\mu_0} C$

Energy Densities of EM waves

$u_E = \frac{1}{2}\varepsilon_0 E^2$

$\langle u_E \rangle = \frac{1}{4}\varepsilon_0 E_0^2 = \frac{1}{2}\varepsilon_0 E_{rms}^2$

$u_T = \frac{1}{2}\varepsilon_0 E_0^2$

$u_B = \frac{B^2}{2\mu_0}$

$\langle u_B \rangle = \frac{B^2}{4\mu_0} = \frac{B_{rms}^2}{2\mu_0}$

$u_T = \frac{B_0^2}{2\mu_0}$

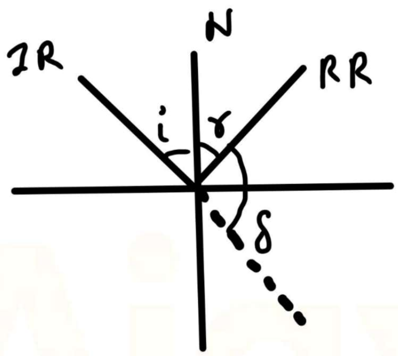
RAY OPTICS

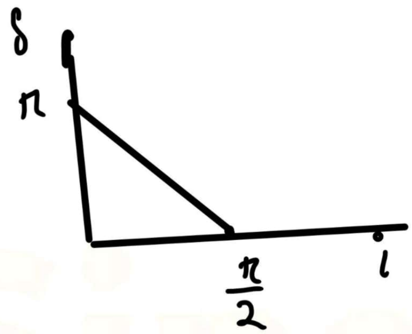
Reflection of Light

IR, RR & N
lies in the same plane

$\angle i = \angle r$

$\delta = 180^\circ - 2i$

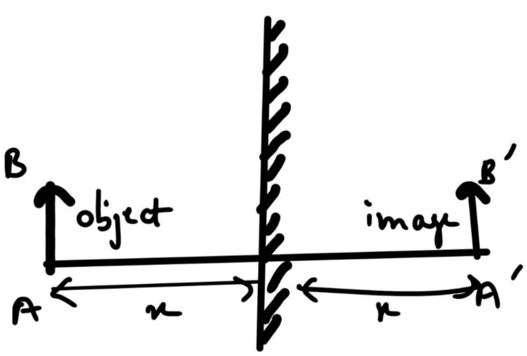


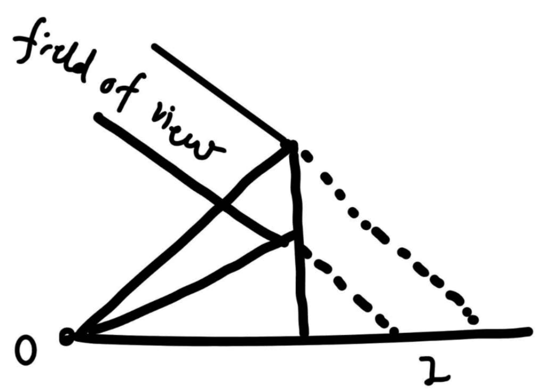


Mirror Formula

$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$ $\boxed{f = \frac{R}{2}}$ $\boxed{v = \frac{uf}{u-f}}$

Properties of Images from plane & spherical mirrors:





Magnification

$$m = \frac{\text{Size of image in transverse plane}}{\text{Size of object in transverse plane}} \qquad m = \pm 1$$

No. of images formed by two inclined mirrors:

If $\boxed{\frac{360^\circ}{\theta} = \text{even}}$ $\Rightarrow$ $\boxed{n = \frac{360^\circ}{\theta} - 1}$

If $\boxed{\frac{360^\circ}{\theta} = \text{odd}}$ $\Rightarrow$ 1) If object is placed on angle bisector

$$\boxed{n = \frac{360^\circ}{\theta} - 1}$$

2) If object is not placed on angle bisector

$$\boxed{n = \frac{360^\circ}{\theta}}$$

Motion of Images

	In transverse plane	In longitudinal plane
If mirror is not moving	$\vec{v_i} = \vec{v_o}$	$\vec{v_i} = -\vec{v_o}$
If mirror is moving	$\vec{v_i} = \vec{v_o}$	$\vec{v_i} = 2v_m - \vec{v_o}$

v_o velocity of object

v_i velocity of image

v_m velocity of mirror